

QUAD CITY, IL, USA

WMO: 725440

Lat: 41.450N

Lon: 90.517W

Elev: 594

StdP: 14.38

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14923

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.3	(c) 1.2	(d) -12.4	(e) 2.9	(f) -3.0	(g) -7.6	(h) 3.7	(i) 2.2	(j) 26.4	(k) 22.7	(l) 24.3	(m) 24.1	(n) 7.8	(o) 280	(p) 0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.4	(c) 92.3	(d) 76.2	(e) 89.8	(f) 74.9	(g) 87.2	(h) 73.2	(i) 79.1	(j) 88.8	(k) 77.3	(l) 86.5	(m) 75.6	(n) 84.2	(o) 10.2	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.3	140.4	85.5	74.5	131.9	83.4	72.8	124.3	81.4	43.0	88.7	41.1	86.5	39.5	84.7	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.9	(b) 20.3	(c) 18.3	(e) -14.2	(f) 96.5	(g) 8.6	(h) 2.8	(i) -20.4	(j) 98.6	(k) -25.4	(l) 100.2	(m) -30.2	(n) 101.8	(o) -36.4	(p) 103.8		
(4) 23.9	(5) 20.3	(6) 18.3	(7) -14.6	(8) 82.6	(9) 8.4	(10) 2.0	(11) -20.6	(12) 84.0	(13) -25.5	(14) 85.2	(15) -30.2	(16) 86.3	(17) -36.3	(18) 87.7		

Monthly Climatic Design Conditions

		Annual													
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	23.1	27.3	39.5	51.5	62.3	71.9	75.4	73.4	66.3	53.9	40.5	28.7
(7)		DBStd	20.52	12.55	12.04	11.47	9.45	8.38	6.44	5.68	5.23	7.86	9.41	10.12	11.36
(8)		HDD50	2980	835	638	365	91	6	0	0	0	1	66	314	664
(9)		HDD65	6040	1300	1055	793	416	155	16	2	3	77	364	734	1125
(10)		CDD50	3442	0	3	38	135	387	656	788	724	491	187	30	3
(11)		CDD65	1028	0	0	2	10	71	222	324	263	117	19	0	0
(12)		CDH74	9788	0	0	20	150	691	2116	3283	2246	1098	182	2	0
(13)		CDH80	3459	0	0	0	29	188	778	1302	752	368	42	0	0
(14)	Wind	WSAvg	8.2	9.3	9.3	9.7	10.3	8.5	7.1	6.2	5.8	6.4	7.9	8.8	8.7
(15)	Precipitation	PrecAvg	38.4	1.6	1.5	2.8	3.9	4.4	4.6	4.6	4.2	3.5	2.9	2.4	2.0
(16)		PrecMax	56.4	4.4	3.5	7.4	11.3	11.4	13.2	11.8	15.2	14.2	8.5	6.8	5.0
(17)		PrecMin	17.8	0.3	0.2	0.6	0.9	0.3	1.0	0.4	0.4	0.0	0.0	0.3	0.0
(18)		PrecStd	7.5	0.9	0.9	1.6	1.9	2.3	2.8	2.6	2.9	2.6	1.9	1.5	1.2
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	56.5	63.3	77.1	83.9	89.7	94.0	97.1	93.5	91.8	85.5	72.2	61.6	(19)
(20)		0.4% MCWB	48.8	53.3	62.8	63.3	71.2	76.2	79.2	76.0	73.9	70.3	58.7	58.5	(20)
(21)		2% DB	49.2	55.4	69.2	78.6	84.9	90.9	92.9	90.1	88.1	79.0	65.7	55.0	(21)
(22)		2% MCWB	44.2	48.5	57.9	62.1	68.7	74.3	77.6	76.2	72.4	64.8	56.0	50.5	(22)
(23)		5% DB	43.2	48.9	63.0	73.2	81.3	87.8	89.9	87.1	84.3	74.2	60.9	48.9	(23)
(24)		5% MCWB	39.3	43.7	54.8	58.8	66.6	72.6	76.1	74.2	69.8	62.5	53.4	44.3	(24)
(25)		10% DB	38.8	43.4	57.2	68.5	77.7	84.7	86.9	84.2	80.6	69.6	56.6	43.6	(25)
(26)		10% MCWB	35.9	39.3	49.4	56.1	64.9	71.0	74.3	72.4	68.0	60.3	50.0	39.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	52.1	56.1	64.6	67.2	74.8	79.2	82.2	80.5	77.1	71.8	62.0	58.9	(27)
(28)		0.4% MCDB	54.2	59.7	74.3	79.0	85.9	89.9	92.5	89.7	87.5	82.4	68.8	61.9	(28)
(29)		2% WB	44.0	49.1	59.9	64.1	71.5	77.0	80.0	78.1	74.6	68.2	57.8	50.7	(29)
(30)		2% MCDB	48.5	54.8	66.9	74.1	81.5	87.2	89.6	86.8	84.3	75.3	63.2	54.0	(30)
(31)		5% WB	39.5	43.7	55.3	61.1	69.2	75.1	78.1	76.3	72.5	64.6	54.5	44.9	(31)
(32)		5% MCDB	43.1	48.4	62.8	71.2	77.8	84.6	87.3	84.1	80.7	71.4	60.2	48.4	(32)
(33)		10% WB	35.9	39.4	49.8	57.8	66.8	73.1	76.4	74.6	70.3	61.5	50.2	39.9	(33)
(34)		10% MCDB	38.6	43.0	57.1	67.0	75.0	81.7	84.8	81.7	78.0	68.9	55.8	43.4	(34)
(35)	Mean Daily Temperature Range	MDBR	15.5	16.2	18.7	21.1	20.8	19.9	19.4	19.2	21.9	20.2	17.3	14.5	(35)
(36)		5% DB MCDBR	20.6	23.3	25.8	28.6	24.3	23.0	21.3	21.3	24.8	25.6	23.9	20.5	(36)
(37)		5% DB MCWBR	16.5	17.3	16.4	16.0	12.2	10.4	9.8	10.2	11.5	14.0	16.4	16.2	(37)
(38)		5% WB MCDBR	19.4	21.6	23.1	23.9	21.4	20.2	19.0	18.7	20.5	20.7	21.3	18.9	(38)
(39)		5% WB MCWBR	16.0	16.9	15.6	15.4	12.5	10.4	10.0	10.2	11.0	13.6	16.5	16.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.305	0.343	0.390	0.424	0.427	0.445	0.426	0.396	0.348	0.304	0.292	(40)
(41)		taud	2.416	2.391	2.377	2.252	2.222	2.266	2.197	2.252	2.307	2.443	2.500	2.436	(41)
(42)		Ebn at Noon	274	287	287	279	271	270	263	265	265	268	268	264	(42)
(43)		Edh at Noon	25	30	34	42	44	43	45	41	36	29	23	23	(43)
(44)	All-Sky Solar Radiation	RadAvg	559	871	1185	1483	1736	1950	1935	1711	1453	957	636	465	(44)
(45)		RadStd	65	89	96	130	126	171	126	113	120	110	54	50	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page